



Standardizing Quantitative Visual Stratigraphy for Digital Archiving of Ice Cores

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Introduction

Ice cores preserve a continuous record of past environmental and climatic conditions through variations in ice structure, chemistry, and physical properties. Visual stratigraphy provides one of the most fundamental datasets collected from an ice core, documenting features such as annual layering, bubble distribution, fractures, dust horizons, and other englacial structures that aid interpretation of complementary measurements including electrical conductivity, stable isotopes, and geochemical analyses.

Despite its importance, standardized high-resolution visual documentation is often incomplete in archived ice cores, limiting their long-term analytical value. While methods such as electrical conductivity measurements (ECM) and hyperspectral imaging (HSI) provide detailed characterization, they require specialized instrumentation not available across all curation facilities.

This study develops a low-cost, reproducible imaging workflow using commercially available digital photography to produce calibrated, spatially continuous visual records of archived ice core AH2416 (Allan Hills Blue Ice Area, Antarctica). The workflow integrates dual-illumination imaging and manual stratigraphic logging to generate paired, depth-referenced visual and descriptive datasets suitable for long-term curation and future quantitative analysis.

Objectives

- Develop a low-cost, reproducible workflow for high-resolution visual stratigraphy.
- Produce a permanent visual archive of archived ice core AH2416.
- Create spatially aligned photographic and manual stratigraphic datasets for future quantitative analyses.

Methods

Digital Photography

A standardized digital imaging workflow was developed to produce a continuous, depth-referenced visual archive of the Allan Hills shallow ice core (AH2416). Individual core sections were photographed under transmitted-light and reflected-light illumination using a DSLR camera mounted on a precision linear rail. Transmitted-light imaging enhanced visualization of internal stratigraphic features and bubble distributions, whereas reflected-light imaging emphasized surface textures and fractures. A Stouffer 21-step transmission wedge was included during transmitted-light imaging to provide a reference for image intensity calibration. Images were acquired at 35-cm intervals using fixed manual camera settings, with adjacent photographs overlapping by 10 cm to facilitate assembly into continuous composite images. Image processing was performed in RawTherapee version 5.12 and Fiji/ImageJ for image intensity calibration and assembly of composite images for visual stratigraphy analysis.

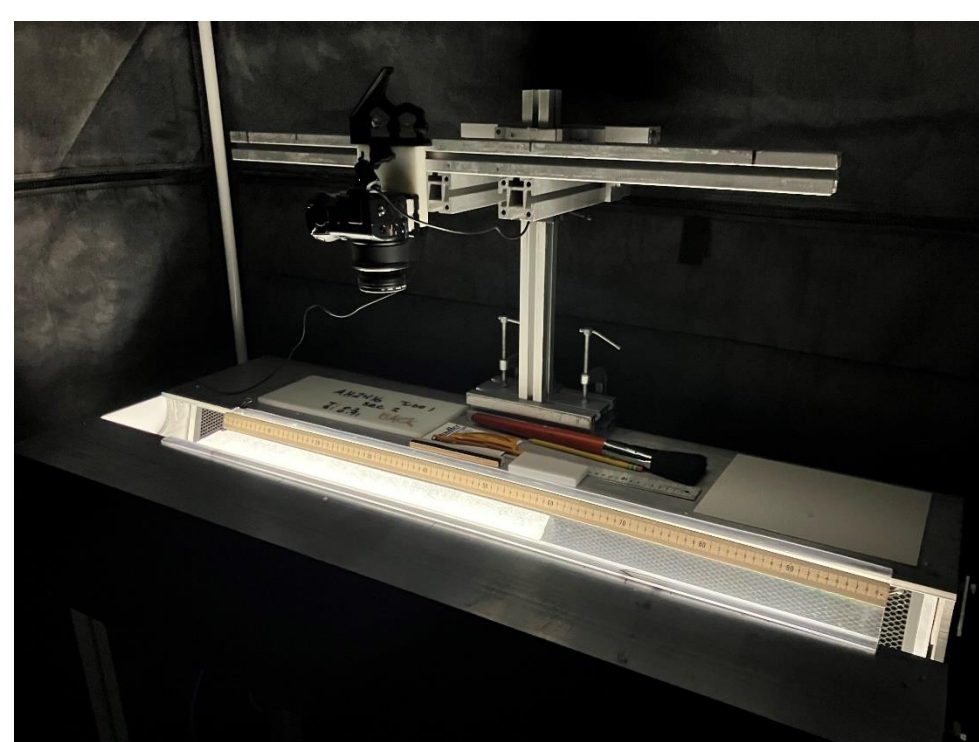


Figure 1.

Imaging setup used for ice core photography. A light box provided uniform backlighting to enhance the visibility of internal ice features. Ice core sections were photographed using an Olympus C-8080 Wide Zoom digital camera mounted on a rail system, which enabled consistent camera positioning and sequential image acquisition for later stitching and analysis.

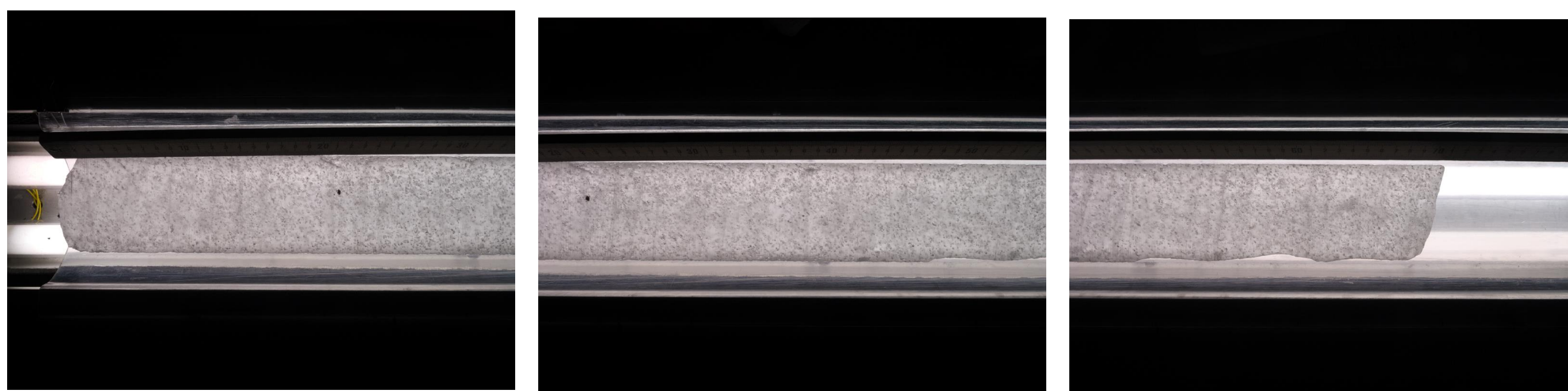


Figure 2.

Unprocessed backlit photographs of Tube 12, Section 13 from ice core AH2416. The images represent the original sequential photographs acquired prior to image preprocessing and stitching, illustrating the overlap between adjacent photographs used to create the final composite image.

Manual Visual Logging

Ice core sections were manually logged at 1:1 scale on graph paper to document visual stratigraphic features at millimeter resolution. Logging followed standard visual core description practices and included observations of tephra layers, bubble density, fractures, dust, rock inclusions, and cloudy bands. Logging was conducted continuously along core depth during processing. Completed logs will be digitized via scanning and archived alongside the photographic dataset for future comparative use.

Results

Backlit imaging and image stitching produced continuous, high-resolution composite images of the ice core. The workflow improved visualization of stratigraphic features, including fractures, air bubbles, rock inclusions, and grain-size variations. Core logs provided complementary millimeter-scale documentation of these features.

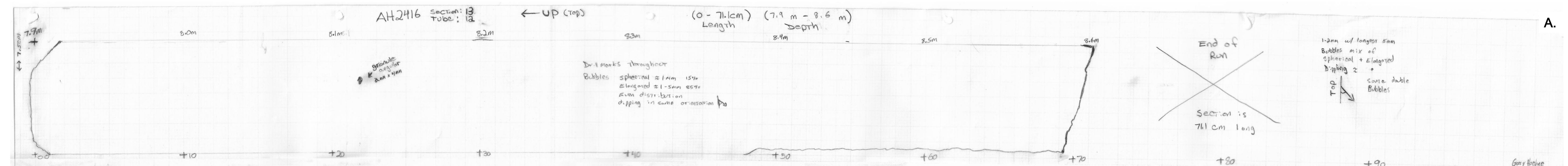


Figure 3.

One-meter core logs were positioned parallel to the ice core on the light table to document stratigraphic features at millimeter-scale resolution. Visual observations, including fractures, air bubbles, rock inclusions, and variations in grain size were recorded directly on the core logs to provide a detailed stratigraphic record for subsequent analysis.

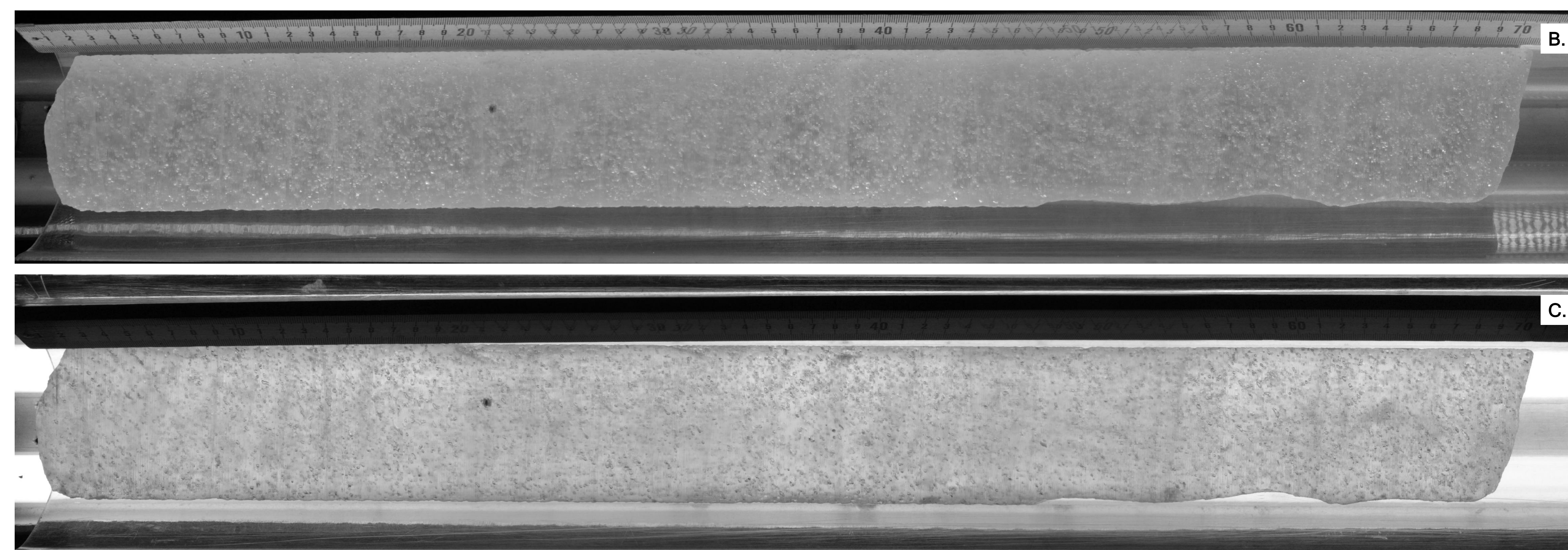


Figure 4.

Comparison of visual logging and stitched ice core mosaics for Tube 12, Section 13 of ice core AH2416. (A) Visual log documenting observed stratigraphic features observed during core examination. (B) Stitched mosaic acquired using reflected (side) illumination, highlighting surface characteristics of the ice core. (C) Stitched mosaic acquired using transmitted (backlit) illumination, enhancing the visibility of internal stratigraphic features, including fractures, air bubbles, and sediment inclusions.

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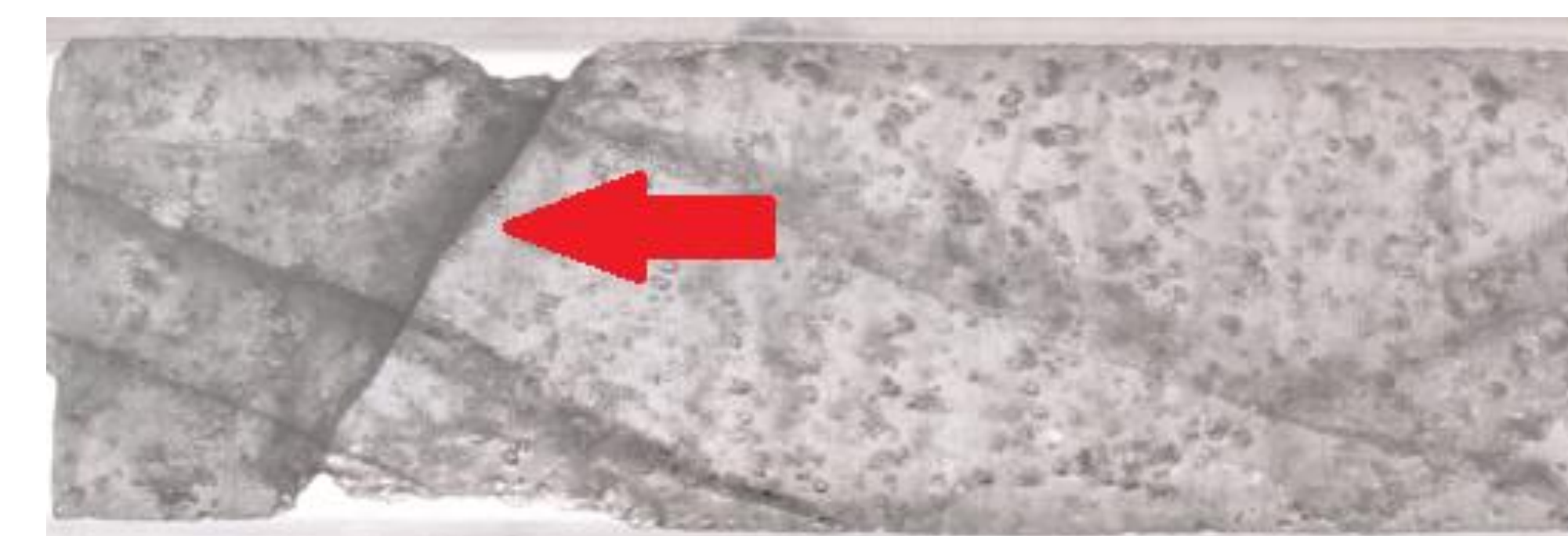
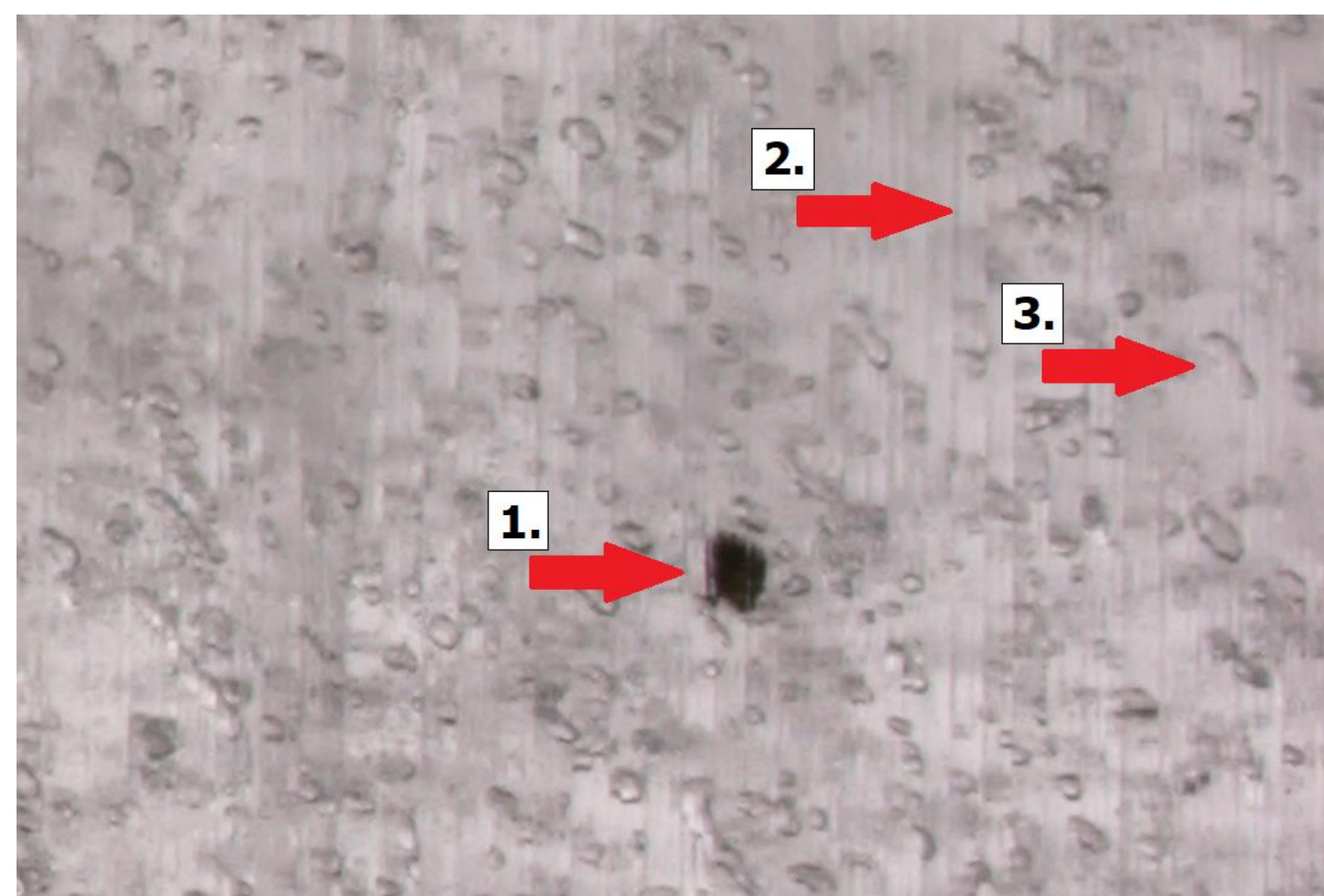


Figure 5.

Representative stratigraphic features observed in the stitched transmitted-light mosaic of ice core AH2416. (A) Annotated features include (1) a rock inclusion, (2) aa drill mark, and (3) an elongated air bubble indicated an apparent dip direction. (B) Fractures observed within the ice core, illustrating structural features visible in transmitted-light imagery.

Conclusion

- A reproducible workflow was developed for imaging and stitching ice core sections.
- Transmitted-light mosaics enhanced the visualization of internal stratigraphic features compared to reflected-light images.
- The combined imaging and visual logging approach provides high-resolution documentation for ice core analysis.

Future Applications

- Quantitative image analysis
- Automated layer detection
- Machine-learning feature recognition
- Comparison with ECM and hyperspectral (HSI) datasets
- Rapid screening of large ice cores sections to identify regions of scientific interest
- Long-term digital preservation of archived ice cores

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